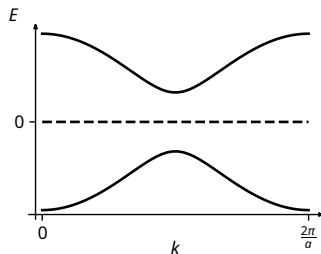
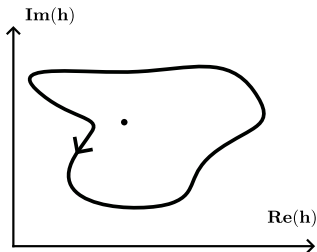


Phase determined by symmetries of the bulk:
bulk-boundary correspondence

Example: chiral symmetry

$$\rightarrow H = \begin{pmatrix} 0 & h(\vec{k}) \\ h^\dagger(\vec{k}) & 0 \end{pmatrix}$$



Topological phases can be distinguished
by topological invariants

in 1D: winding number